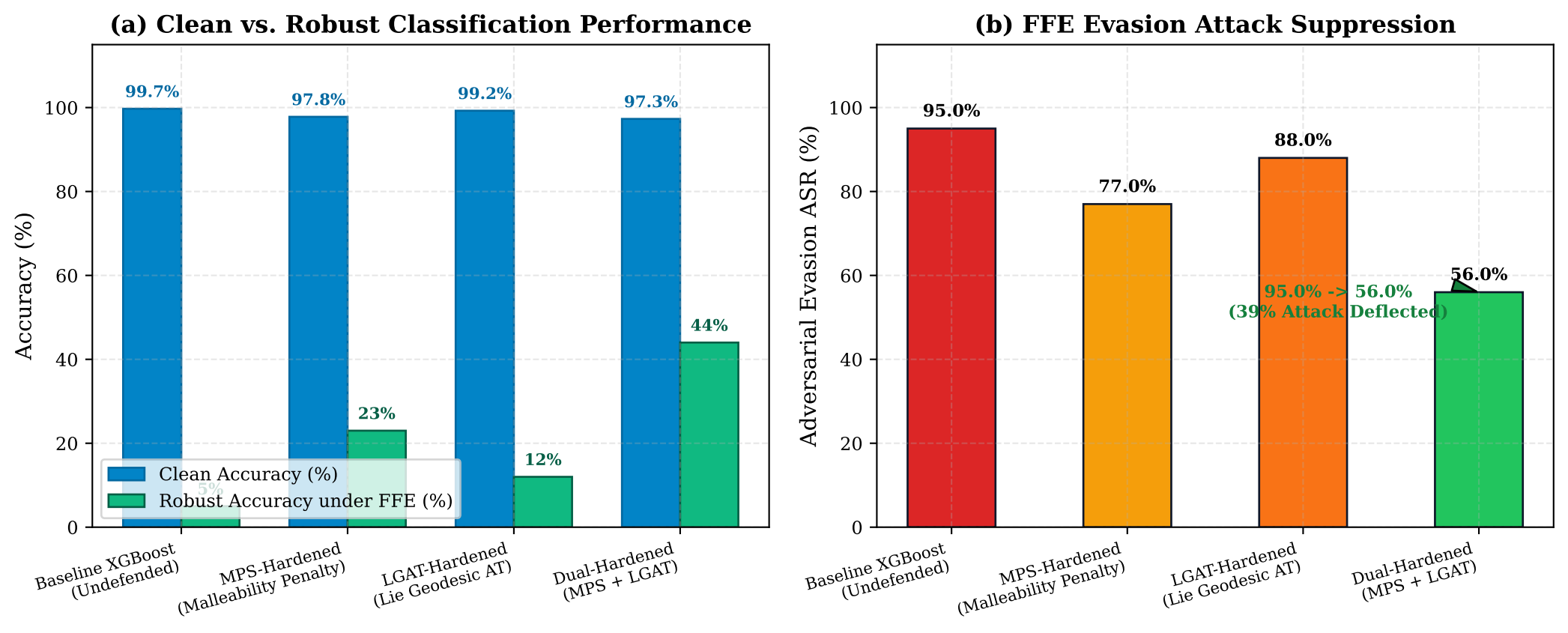
(a) Clean vs. Robust Classification Performance
Accuracy (%)
0
20
40
60
80
100
99.7%
97.8%
99.2%
97.3%
5%
23%
12%
44%
Clean Accuracy (%)
Robust Accuracy under FFE (%)
Baseline XGBoost (Undefended)
MPS-Hardened (Malleability Penalty)
LGAT-Hardened (Lie Geodesic AT)
Dual-Hardened (MPS + LGAT)
(b) FFE Evasion Attack Suppression
Adversarial Evasion ASR (%)
0
20
40
60
80
100
95.0%
77.0%
88.0%
56.0%
95.0% -> 56.0%
(39% Attack Deflected)
Baseline XGBoost (Undefended)
MPS-Hardened (Malleability Penalty)
LGAT-Hardened (Lie Geodesic AT)
Dual-Hardened (MPS + LGAT)